

WAYBILL: Receiver-Sealed Origin Labels and Odometer-Backed Settlement for Disclosure-Minimizing Road-Usage Charging

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Abstract—Disclosure-minimizing road-usage charging can avoid disclosing raw trajectories by computing fees locally and proving correct payment. Correctness alone does not establish completeness when a vehicle can label an interval as unavailable or withhold a whole charging period. We present WAYBILL, a conditional settlement design that assigns acquisition-label admissibility to a process-isolated receiver signer while keeping tariff selection and bill computation outside that trusted process. The signer seals acquisition-supplied fix records, monotone odometer readings, and an admissibility bit derived from each origin-fix status. The vehicle proves in zero knowledge that every reported interval follows a committed tariff or an odometer-backed maximum-rate fallback. At month close, signed odometer distance not covered by accepted non-overlapping periods is reconciled at the same maximum rate. We prove that fallback and period withholding cannot reduce the bill relative to acquisition-supplied, receiver-sealed tariff labels, conditional on correct receiver attestations, proof soundness, and monthly odometry. The guarantee does not cover receiver compromise, odometer tampering, or GNSS relay. A receipted 25-fix Groth16 run of the V6 prototype contains 246,255 constraints and 22 public signals. It produced an 803-byte JSON proof encoding in 3.65 s and verified in 217 ms with exact circuit and reference agreement. Endpoint receipts further record replay rejection, public-signal tamper rejection, and month-close reconciliation. These measurements validate one canonical execution path. The pre-registered four-dataset trace evaluation of billing accuracy, scaling, bounded displacement, metadata linkage, and handler behavior remains pending, and no full-evaluation claim is made from the canonical receipt.

Index Terms—road-usage charging, location privacy, zero-knowledge proofs, trusted sensing, accountable settlement, GNSS

I. INTRODUCTION

Road-usage charging (RUC) can price distance, place, and time more directly than a fuel tax. A direct Global Navigation Satellite System (GNSS) design, however, gives the operator a detailed movement history. Private tolling protocols instead compute charges locally and use commitments, proofs, or audits to establish correct payment [1], [2], [3], [4]. This line of work has made fee correctness increasingly precise, yet a distinct systems problem remains. A correct proof covers the records it contains. It does not by itself show that every chargeable interval or period was submitted.

Completeness is commonly supported by infrastructure observations such as gates, cameras, or mobile checks. A recent survey organizes these privacy and enforcement tradeoffs, while a deployment case study shows that protocol-level

privacy alone need not remove linkability from the surrounding collection infrastructure [5], [6]. The problem is therefore broader than hiding coordinates inside a proof. A settlement design must bind who decides whether location data are usable, account for periods that never reach the verifier, and disclose the remaining account-level metadata explicitly.

WAYBILL studies this problem for vehicles that already contain an enrolled sensing unit and a calibrated monotone odometer. Its central design choice is to separate origin-label admissibility from billing. A process-isolated receiver signer seals fixes, odometer readings, and origin-label admissibility decisions. The normal vehicle process proves tariff membership and bill arithmetic, while the charger checks non-overlap and reconciles unreported monthly distance. This boundary remains conditional. Navigation-message authentication can authenticate transmitted navigation data, but it does not prove that a receiver occupied a physical location or defeat every relay attack. Galileo Open Service Navigation Message Authentication (OSNMA) status is consequently an acquisition input to WAYBILL, not a complete location oracle [7], [8], [9].

The resulting argument is narrow but operationally useful. Receiver-sealed validity prevents a normal prover from choosing a cheaper outage path, and odometer-backed month close removes the discount from withholding a period under stated sensing assumptions. The contributions are as follows.

- 1) We define a validity-bound settlement interface in which a process-isolated receiver signer derives and commits origin-label admissibility with each fix. The normal prover cannot replace fallback bits, roots, tariffs, policy profiles, or verification keys after acquisition.
- 2) We combine interval fallback with odometer-backed month accounting and prove a conditional no-discount result. Relative to acquisition-supplied, receiver-sealed tariff labels, neither declaring fallback nor withholding complete periods can lower the charge when the receiver, odometer, and proof assumptions hold.
- 3) We implement the end-to-end V6 path and pre-register a five-stage evaluation over four mobility datasets. Immutable receipts bind code, data, container, policy, and artifacts. This draft reports only the canonical differential receipt and reserves all full-dataset claims for the strict server aggregate.

These contributions separate relation correctness, conditional completeness, and disclosure minimization. We make no universal deployment claim, no claim of physical location truth, and no information-theoretic route-privacy claim.

II. RELATED WORK

A. Private tolling and completeness

VPriv computes location-based fees while limiting disclosure through local computation, commitments, and audits [1]. PrETP embeds local fee computation in an optimistic payment protocol and uses external evidence to challenge committed positions [2]. The Phantom Tollbooth addresses collusion and conceals spot-check locations during audits [3]. P4TC develops provable privacy and security properties for practical DSRC toll collection [4]. The broader literature spans centralized, distributed, and hybrid designs with different observation and trust assumptions [5]. PriPAYD similarly demonstrates the relevance of local computation and trusted on-board usage data outside toll collection [10]. WAYBILL does not subsume these protocols. It asks whether an enrolled receiver and monthly odometry can replace part of the recurring observation function in a GNSS setting, at the cost of a stronger trusted-sensing boundary.

The closest abstraction is a maximum-rate deposit followed by verified discounts, which already gives the theorem’s inequality. Our narrower systems contribution is to make discount conditions auditable through receiver sealing, proof binding, non-overlap, and an append-only month ledger. Attested on-board meters or random roadside checks offer different assurance/availability tradeoffs; we do not claim process isolation dominates them.

B. Protocol privacy and deployment leakage

Cryptographic privacy can be weakened by identifiers, account workflows, receipts, and exception handling outside the core protocol. The Brisbane case study demonstrates why the surrounding collection infrastructure belongs in the privacy analysis [6]. WAYBILL follows this lesson by declaring the complete public settlement tuple and evaluating linkage under several release views. It does not equate hidden coordinates with an anonymous transaction.

C. Authenticated navigation and zero-knowledge location

Galileo OSNMA authenticates navigation messages received through the Galileo Open Service [7]. It does not establish the physical presence of a receiver, so WAYBILL treats relay and meaconing as residual sensing risks. Location-verification and GNSS-spoofing research make the separate physical-presence problem explicit [9], [8]. Recent zero-knowledge proof-of-location work studies territorial compliance for vehicle subsidies and taxation [11]. Our relation instead binds a sequence of receiver-sealed acquisition records to tariff membership, integer billing, a period ledger, and month-close settlement. These systems solve related but different tasks, and we make no superiority claim.

III. SYSTEM AND THREAT MODEL

A. Parties and trust boundary

The *tariff authority* publishes a versioned cell-to-zone table, zone rates, a maximum rate $r_{\max}$, and a policy profile. The *receiver signer* is a process isolated from the normal prover. Here “independent” denotes process and credential separation, not independent hardware, operator, or sensor verification. An authorized acquisition path appends fixes with a strict status enum. The signer, rather than the caller, derives each outgoing interval bit under the profile-pinned `origin-osnma-authenticated-v1` rule. The *prover* reads sealed period material and constructs a proof. The *charger* enrolls device public keys, pins policy profiles and verification keys, accepts period statements, and closes monthly ledgers.

For each fix f_i , the receiver log contains a device and period identifier, sequence number, authenticated GNSS time, cell label, odometer reading, nonce, and device signature. For every nonterminal fix it also records $v_i \in \{0, 1\}$, indicating whether the profile admits the origin cell label for pricing the next interval. This policy bit is not a physical-position or continued-presence assertion. The normal prover has no API field with which to replace v_i or submit its own root. The receiver signer rejects sequence, time, odometer, and log-epoch rollback. It also rejects duplicate period sealing and a log whose bytes no longer match its stored hash. Before a new seal becomes durable, the signer compares its validated local tail with an authenticated charger head and atomically extends that per-device chain by one epoch. A stale snapshot, deleted tail, epoch gap, or conflicting same-epoch attestation is rejected.

The trusted base includes the enrolled key and acquisition logic. That logic supplies GNSS time, cell label, status, odometer reading, nonce, and monthly odometer boundaries; the signer validates syntax, ordering, profile rules, and chain continuity but does not parse OSNMA messages or independently recompute a cell from a navigation solution. It also assumes odometer authenticity, monotonicity, calibration accuracy, and continuity across maintenance. These facts are software-signed, not remote or hardware attestations. The trusted base excludes tariff selection and bill arithmetic. We do not address key extraction, a maliciously enrolled receiver, undetectable physical odometer tampering, ECU/CAN compromise, systematic under-recording, or a compromised acquisition process; remote automotive attack surfaces motivate this exclusion [12].

B. Adversary and goals

An adversarial prover may alter private witness values, substitute a tariff or verification key, replay an old proof, rebind a statement to another period or month, omit intervals or entire periods, or modify public signals. It may not forge the receiver signature or the proof system. Signal relay and meaconing are evaluated as residual sensing attacks, not ruled out cryptographically.

An accepted period should have the fee, distance, fallback count, roots, profile commitment, and canonical time windows proved together. A withheld period should not create an economic discount under the conditional month-close model. The privacy goal is narrower. Raw fixes and interval charges are absent from the proof-only request. The charger still learns the device-month link, fee, distance, number of fallback intervals, time bounds, and commitments.

IV. WAYBILL DESIGN

A. Design overview

WAYBILL receives a sealed sequence of fixes for one charging period and produces two linked outputs. The first is a zero-knowledge statement containing the period totals and commitments required for verification. The second is an immutable receipt that binds the accepted statement to receiver and charger state. Four mechanisms form the settlement chain. A validity-bound receiver root fixes which intervals may use ordinary tariff labels. A period relation proves tariff membership and arithmetic. A verifier transition rejects replay and overlap. Monthly reconciliation charges odometer distance absent from all accepted periods. Figure 1 shows how responsibility passes between these mechanisms.

B. Binding origin-label admissibility

Let H denote the circuit's Poseidon chaining construction. For a nonterminal fix, the receiver and circuit compute

$$c_i = H(\text{DAC}, \text{profile}, \text{period}, i, t_i, x_i, y_i, o_i, n_i, v_i).$$

where DAC identifies the data-acquisition contract, profile and period bind the policy and period, $(t_i, x_i, y_i, o_i, n_i)$ are the signed time, cell, odometer, and nonce fields, and v_i is the profile-derived origin-label admissibility bit. Here (x_i, y_i) are discrete cell labels, not receiver-signed sub-cell physical coordinates. The final fix uses the same tuple with a fixed false sentinel. Chaining all c_i yields R_{fix} . Consequently, changing a validity decision changes the root used by both the proof and the receiver's Ed25519 software attestation. This removes an interface in which signatures could cover fixes while leaving fallback flags under prover control.

The signer attests the metadata tuple

$$(\text{receiver}, \text{device}, \text{period}, \text{epoch}, \text{count}, R_{\text{fix}}, \\ \text{DAC}, \text{semantics}, \text{rule}, \text{charger}, \text{loghash}, \text{prevhash}).$$

The attestation does not sign a statement supplied by the prover. The charger first verifies the V5 attestation signature and appends its hash only if the epoch and previous hash extend the current device head. At settlement, it compares the root, device, period, and device-attestation commitment to the proved public statement and accepts only the exact previously anchored attestation. The read-only preflight avoids spending proof-verification work on an unanchored request. The write transaction repeats the check to preserve correctness under concurrency.

C. Proving the period relation

For interval i , let $\Delta o_i = o_{i+1} - o_i$. The circuit enforces odometer and time monotonicity, a maximum time gap, an odometer speed bound, Boolean v_i , tariff Merkle membership, integer width limits, and the profile-pinned maximum rate. Its fee is

$$q_i = \begin{cases} \Delta o_i r(z_i), & v_i = 1, \\ \Delta o_i r_{\max}, & v_i = 0. \end{cases}$$

where Δo_i is the nonnegative odometer increment, z_i is the acquisition-supplied, receiver-sealed tariff cell, $r(z_i)$ is its committed rate, and $r_{\max}$ is the maximum rate fixed by the policy profile. A signer-sealed tariff cell is therefore an acquisition-supplied label, not independent physical-position verification. Thus a stationary vehicle pays zero even during an outage. The fallback uses the same odometer delta as normal settlement. All products and totals are range-checked under the frozen uint32/uint64/uint96 contract.

The public vector contains 22 fields that bind the fix, tariff, and interval roots, the policy-profile commitment, the period and month identifiers and bounds, the tariff version and device commitment, the totals, the fallback count, and the physical and policy bounds. The proof-only endpoint uses the profile commitment to resolve one authority-approved circuit, tariff root, and verification-key hash. Extra raw fix fields are rejected by the request schema.

D. Reconciling a month

Accepted windows are non-overlapping for each device. At month close, the receiver signs boundary readings $(o_{M,0}, o_{M,1})$. If accepted periods cover distance D_A , the charger computes

$$D_U = \max(0, o_{M,1} - o_{M,0} - D_A), \quad Q_U = D_U r_{\max}.$$

where $o_{M,0}$ and $o_{M,1}$ are the signed month-boundary odometer readings, D_A is the distance in accepted non-overlapping periods, D_U is the remaining distance, and Q_U is its reconciliation charge. An inconsistent excess D_A is rejected outside a named tolerance. A missing month attestation creates a fail-closed administrative exception rather than a distance estimate. Collection policy for that exception is outside the cryptographic theorem; it must not be treated automatically as fraud. Before local persistence, the signer submits the exact signed boundary to the authenticated charger. The first enrolled boundary is a bootstrap trust point. Thereafter the charger accepts only the next calendar month with $o_{M,0}$ equal to the preceding $o_{M-1,1}$. Exact retries are idempotent, while different bytes for an anchored device-month are rejected. This closes stale-local-database rollback at the cost of making charger availability a prerequisite for sealing a month.

E. Verifier state transition

For each statement, the charger resolves the pinned profile, tariff root, circuit, and verification key. It rejects an invalid or unanchored receiver attestation, any mismatch with the exact public vector, a repeated device-period key, or an overlapping

Acquisition and settlement flow. (1) An authorized acquisition component sends unsigned fixes with strict status labels to a device-scoped receiver signer. (2) The signer validates monotonicity, derives v_i from the origin-fix status, and commits each f_i together with v_i or a false terminal sentinel. It seals the append-only log, signs the root metadata, and obtains an exact compare-and-swap confirmation from the charger before committing the local row. (3) The vehicle prover obtains the sealed fixes, admissibility vector, and root attestation, then computes tariff membership, fees, and a Groth16 proof. (4) The charger verifies the profile-pinned proof, exact public-signal vector, receiver attestation, period uniqueness, and non-overlap. (5) Month close bills attested odometer distance not covered by accepted periods at $r_{\max}$.

Fig. 1. The receiver owns the status-derived admissibility decision and root. The normal prover owns neither. Pricing remains outside the receiver process.

window. Only after all checks pass does a transaction record the period and its immutable receipt. When the month is ready to close, the charger validates the boundary attestation against the anchored month chain, computes D_U and Q_U , and records a close receipt. An invalid month boundary creates an administrative exception. No failed check produces a billable period or advances the month ledger.

The prototype does not implement adjudication, but a deployable ledger needs an append-only dispute state machine: $\text{accepted} \rightarrow \text{notified} \rightarrow \text{contested} \rightarrow \text{evidenceOpened} \rightarrow \text{adjudicated} \rightarrow \{\text{adjusted}, \text{closed}\}$. A timely contest should suspend penalties on the disputed amount. Evidence opens in stages—accounting tuple, then endpoints, and raw trajectory only when necessary—under an independent appeals authority. A correction appends a signed adjustment or refund receipt and never overwrites the original. Deadlines, burden of proof, refund interest, and access restrictions are policy obligations rather than claims of the current implementation.

V. SCOPED SECURITY ANALYSIS

Definition 1 (Sealed-label reference bill). For the subset of month distance being compared, associate each odometer distance d_i with a counterfactual acquisition-accepted tariff label z_i and rate $0 \leq r(z_i) \leq r_{\max}$. The reference bill is $Q_{\text{auth}} = \sum_i d_i r(z_i)$. This definition concerns labels accepted by the receiver acquisition boundary. Here d_i is an odometer distance, z_i is its acquisition-accepted tariff label, and $r(z_i)$ is the committed rate. The definition is not a physical-location oracle. A genuinely unavailable interval for which no such reference label exists is outside this comparison and is charged according to the fallback policy without a claim about its unobserved zone rate.

Theorem 1 (Successful-close conditional economic conservatism). *Assume (i) receiver root and month-boundary attestations are unforgeable, (ii) accepted V6 proofs are sound and profile-pinned, (iii) monthly odometer distance is correct and covers the considered month, and (iv) accepted periods are non-overlapping and their accounted distance is subtracted exactly once. Then replacing any reference-labeled interval by fallback, or withholding any set of period statements and closing the month, cannot reduce the bill below Q_{auth} . The inequality is strict for positive withheld distance whose sealed-label rate is below $r_{\max}$.*

Proof. For an accepted interval with $v_i = 1$, the circuit charges $d_i r(z_i)$. With $v_i = 0$, it charges $d_i r_{\max} \geq d_i r(z_i)$. Every

distance not represented by a non-overlapping accepted period remains in D_U and is charged at $r_{\max}$, again no less than its reference rate. Summing these disjoint contributions proves the inequality. Strictness follows when $d_i > 0$ and $r(z_i) < r_{\max}$ for at least one replaced contribution. $\square$

Equivalently, after a successful month close the mechanism starts from charging all covered month distance at $r_{\max}$ and subtracts only the verified discounts $d_i(r_{\max} - r(z_i))$ for accepted ordinary-rate intervals. The algebraic monotonicity is immediate; the systems contribution is binding those discounts to sealed labels, non-overlapping periods, and append-only state. The theorem makes no claim for an indefinitely missing month attestation, which remains an administrative exception requiring the dispute process above.

Theorem 1 is intentionally not called “physical tariff truth.” If a compromised receiver signs a false cell, the proof faithfully bills that false sealed label. If odometry understates distance, month close inherits the error. OSNMA authenticates navigation messages, whereas the receiver’s actual presence and resistance to delayed relay require additional sensing and operational controls. These are assumption failures, not attacks solved by the settlement circuit.

a) Proof and replay binding: The charger compares the full public vector against the submitted statement, uses only the profile-pinned verification key, checks the validity-bound root attestation, rejects duplicate (device, period) keys, and rejects overlapping device windows. A valid proof detached from its original statement therefore fails signal equality or profile resolution.

b) Privacy boundary: We do not provide a simulation proof for an application-level privacy notion. Instead, we declare the charger view and measure how uniquely it identifies a period in a candidate pool. Hash commitments hide neither low-entropy public totals nor linkability induced by the billing account. Bucketed release is a policy mitigation with an explicit accuracy/leakage tradeoff.

VI. IMPLEMENTATION

A. Prototype and receipt discipline

The V6 reference is implemented in Python and Circom. Its commitment and Merkle constructions use the circuit-oriented Poseidon hash [13]. Groth16 is used for its small proofs and mature tooling, not as a novel proof-system contribution [14]. The canonical policy is `ruc-demo-v9`. It pins tariff `v9`, the validity-bound 25-fix circuit, maximum rate, integer

TABLE I
CANONICAL VALIDITY-BOUND V6 ARTIFACT RECEIPT.

Metric	Receipt value
Fixes / intervals	25 / 24
Constraints	246,255
Public signals	22
Fallback intervals	3
Witness generation	0.408 s
snarkjs proving	3.646 s
snarkjs verification	0.217 s
Proof JSON	803 bytes
Circuit/reference vector	exact match

semantics, and verification key. The checked-in direct setup is explicitly test-only. Production requires an independently governed ceremony or a different setup strategy.

The receiver signer is a device-scoped FastAPI process backed by owner-only SQLite state. Its acquisition route requires a write bearer token, accepts only unsigned measurement fields, derives rather than accepts the origin-label admissibility bits, creates the receiver signatures itself, and seals each period and monthly odometer boundary once. A separate read bearer token protects prover access to sealed trajectories. Root-attestation V5 pins the validity rule and charger domain and includes the previous attestation hash. The signer uses an authenticated TLS channel to extend a Charger-side compare-and-swap head before persisting the local row. An exact one-link-ahead retry recovers a lost response or local commit. Monthly-attestation V2 also binds the intended charger domain and is online-anchored under exact-retry and conflict-rejection semantics. The signer pins the canonical profile itself, enforces its validity window and fix cap, and opens its key, database, and profile with fail-closed ownership and mode checks. Its SQLite schema is versioned and rejects injected triggers, views, unknown objects, and unsupported legacy layouts. The charger stores enrollments, accepted periods, month state, immutable receipts, and audit events transactionally. The formal server configuration uses PostgreSQL, administrator and device bearer credentials, and one application worker. SQLite remains a local-test backend. Formal stage-result V2 records include the job hash and recursive artifact manifest. Attempt and final gate receipts are hashed. Merge revalidates files from disk rather than trusting a copied `passed` value.

VII. EVALUATION METHODOLOGY AND CURRENT EVIDENCE

A. Canonical-path validation

Table I reports the checked-in receipt generated on an arm64 development host with snarkjs. Timing comes from one artifact-validation run and is not a population estimate. The receipt explicitly records the test-only setup.

The same proof passed an in-process endpoint-logic gate using FastAPI TestClient and SQLite test mode; this is not deployed server evidence. Re-submission was rejected with

HTTP 409. Changing the public fee signal was rejected with HTTP 400. The accepted period covered 2,400 m and 3,600 cents. A signed 3,000 m month boundary caused exactly 600 m to be reconciled at five experimental cents/m, for a 3,000-cent supplement. The endpoint receipt also records that its root attestation came from a separate receiver process, names the sealed-log SHA-256, and records the exact V5 chain head anchored before proof acceptance. It separately binds the monthly attestation SHA-256, confirms exact-retry idempotence, and records rejection of a conflicting validly signed boundary for the same device-month.

B. Research questions and pre-registered protocol

The full evaluation is organized around five research questions. These trace datasets contain neither real OSNMA verification states nor independently measured vehicle odometers; the experiment is mechanism sensitivity, not a certification of receiver correctness, anti-tamper odometry, or physical location truth. RQ1 asks whether V6 preserves trace-derived reference billing accuracy under synthetic independent and bursty unavailability, synthetic odometer error, and whole-period withholding. RQ2 measures how circuit depth and period size affect compilation, setup, proving, verification, memory, and JSON proof-encoding size. RQ3 quantifies the billing effect of bounded cell/trajectory displacement across tariff granularities; it is not a signal-level relay or meaconing experiment. RQ4 measures identity linkage under exact, fallback-augmented, bucketed, and selectively opened public views. RQ5 tests end-to-end CLI proof verification and the in-process FastAPI handler/SQLite ledger under valid, tampered, and replayed concurrent workloads. It does not measure HTTPS/Uvicorn/PostgreSQL deployment capacity.

The protocol uses T-Drive, GeoLife, Porto, and Rome mobility traces. It applies five fixed seeds and does not cap trips, raw points, inputs, or fixes. S1 compares a trace-derived position reference, the earlier V5 relation, V6, and V6 without month close. The earlier private tolling systems provide design baselines for local computation, commitment auditing, and spot-check enforcement [1], [2], [3]. They are not relabeled as executable V6 baselines because their trust and observation models differ.

Formal jobs will run in a frozen Linux x86-64 container with Python 3.12, Circom 2.1.9, snarkjs 0.7.6, and the policy-pinned artifacts. Each setup or proving job receives eight CPU cores and 32 GiB of memory. The largest handler stage receives 32 CPU cores and 64 GiB. The target starts with at least 500 GiB of free workspace and the exact 4.83 GB powers-of-tau file. These are scheduler allocations and admission thresholds. Processor model and operating-system details will be read from the target-host receipt after the server run.

S1 will report vehicle or user cluster-bootstrap 95% confidence intervals with 5,000 replicates and treats every rejected period as fail closed. Its month-close ablation removes the complete close mechanism. S2 will record warm-up trials separately and report the median, 95th percentile, and 99th

TABLE II
PRE-REGISTERED EVALUATION STAGES AND CURRENT EVIDENCE
BOUNDARY.

Stage	Primary evidence	Draft status
S1	Billing accuracy and completeness	Server aggregate pending
S2	Proof cost and scaling	Canonical V6 receipt only
S3	Bounded displacement sensitivity	Server aggregate pending
S4	Metadata linkage	Server aggregate pending
S5	CLI verifier and in-process handler	Server aggregate pending

percentile for measured trials. With ten trials, P95 and P99 are descriptive nearest-rank maxima rather than stable tail estimates. S3 freezes four tariff bucket sizes and three bounded-displacement radii. S4 separates four disclosure views. These are exact totals, exact totals with fallback metadata, bucket-only totals, and bucket-only totals with opened cells. Bucket views contain no duplicate exact totals. Gallery and query training and test identities are disjoint. Gallery and query records use the same held-out identities but disjoint periods, and a one-period temporal gap is enforced. S5 will use distinct real Groth16 bundles rather than repeatedly verifying one cached proof.

These choices avoid two evaluation shortcuts. A bootstrap over rows would overstate precision when periods from one driver are correlated, so the resampling unit is the original vehicle or user. Retaining exact totals beside bucketed values would make a bucketed treatment observationally identical to the exact view, so each view has its own strict schema.

C. Evidence boundary

Table I is the only quantitative proof-performance result claimed before the formal run. The four-dataset S1–S5 results will be inserted only from the strict aggregate after every expected job has a successful immutable attempt. A time-out, out-of-memory event, rejected period, or missing dataset is retained in `hash-bound failure-summary.json` and `scientific-aggregate.json`. The latter binds selected attempts, parameters, stage metrics, per-seed rows, and median/min/max cross-seed summaries. No result is replaced by a smaller experiment or copied manually from a log.

VIII. DISCUSSION

The contribution of WAYBILL is a settlement boundary rather than a new claim that cryptography can establish physical location. Earlier private tolling systems combine local fee computation with external observations or protocol-specific audits. WAYBILL instead assigns three kinds of evidence to three mechanisms. The receiver seals acquisition records and profile-defined label admissibility, the proof system attests pricing arithmetic, and the monthly odometer attests total distance. This decomposition makes withholding economically conservative under the theorem’s assumptions while preserving

a proof-only, but account-linkable, interface for ordinary settlement. It relocates tariff selection and bill arithmetic outside the acquisition boundary, but does not by itself reduce the acquisition component’s sensing privilege. Whether this trade is preferable to roadside observation depends on deployment governance, hardware assurance, and the acceptable metadata view.

A. Residual sensing and metadata risk

Earlier trajectory sweeps are retained as versioned, non-V6 empirical context, not merged into the canonical cryptographic receipt. They show why the theorem is scoped. Proof-consistent relay can preserve a nonzero billing residual, and exact fee–distance–time tuples can have an anonymity set of one in small pools. Bucketed totals improve this empirical anonymity measure, while opening endpoint fixes during a dispute can sharply reduce it. Consequently, deployment should combine signal-diversity controls, conservative fallback, bucketed public release where policy permits, and accounting-level dispute openings before raw endpoint disclosure.

B. Limitations

The independent signer is a process boundary, not a remotely attested hardware implementation. Separate acquisition and prover tokens reduce privilege, but a compromised acquisition component can still falsify a status label or fix. The derivation rule does not turn software input into hardware truth. Token and key management require rotation, hardware protection, and audit. Online Charger anchoring detects local database rollback, but it does not protect a stolen receiver private key or a host already compromised under the signer UID. A production deployment should move signing and monotonic state into attested hardware. The PostgreSQL charger path has transaction/restart/concurrency tests, but the paper does not claim multi-host failover or multi-worker throughput before target-host stress evidence exists. The canonical proof uses a test-only Groth16 setup. The 25-fix profile is an artifact-scale configuration, not evidence of month-scale client throughput. The current theorem assumes a correct odometer and a completed month attestation. Administrative handling of missing month close remains a governance problem. Finally, the public settlement tuple is linkable and potentially identifying, and we provide an empirical leakage analysis rather than a composable privacy proof.

C. Operational and ethical implications

a) *Billing errors and recourse:* A private bill is not automatically a fair bill. Receiver faults, tariff-map errors, or conservative fallback can overcharge an honest driver. A deployment therefore needs notice, bounded retention, an appeal deadline, and staged dispute openings. It should disclose accounting tuples first and raw endpoints only when necessary. Administrative handling of a missing month attestation must be defined by public policy and cannot be silently interpreted as evidence of fraud. Neither the prototype nor Theorem 1 authorizes an automatic penalty.

M4 cannot establish distributional fairness because its traces lack reliable social and device-quality strata. Deployment work should report false-invalid overcharges, affected-account share, longest fallback run, and correction delay across difficult environments; disputed maximum-rate amounts should remain provisional and receive signed refunds after a confirmed fault.

Governance also lies outside the prototype: tariff authorities own rates/maps, vendors own signer updates/keys, operators own availability/retention, drivers receive notice and contest, and a separate authority audits and adjudicates. Process separation alone does not create organizational independence.

b) Availability: The rollback defense deliberately makes the Charger an online dependency for new seals. A network, TLS, or database outage pauses sealing rather than falling back to an unanchored local signature. Acquisition may retry the exact period after recovery. A timeout after remote commit is recoverable only when the same bytes reproduce the one-link-ahead head. This fail-closed choice prevents silent forking but creates an operational availability and denial-of-service risk.

c) Data governance: Mobility traces are sensitive even when pseudonymized. The four-dataset formal protocol records provenance and hashes, but restricted raw records are not distributed with the repository. Published aggregates must satisfy provider terms and exclude row-level trajectories, device keys, receiver logs, and selective-disclosure payloads. An operational charger should minimize retention of linkable public statements after billing and appeal obligations expire.

d) Artifact availability: The repository includes the V6 circuit and wrapper, Python reference, receiver-signer and charger services, policy/tariff artifacts, verification key, tests, formal orchestrator, V2 receipts, and the small canonical proof bundle. Large RICS/WASM/zkey files, the test PTAU, and restricted datasets are referenced by immutable hashes and distributed or mounted separately. The direct Groth16 setup is marked test-only and is insufficient evidence of a production ceremony. The artifact guide gives focused and full verification commands and distinguishes the cryptographic receipt from unavailable formal dataset stages.

IX. CONCLUSION

WAYBILL demonstrates a precise separation between receiver-sealed origin-label admissibility and publicly verified pricing. Binding that bit into the sealed root removes the prover's ability to self-declare an outage, while odometer-backed fallback and month close give a conditional no-discount result for acquisition-supplied, receiver-sealed labels. The result can reduce raw trajectory disclosure and some roadside dependence, but only under explicit sensing and odometry assumptions. Version-pinned artifacts, process-isolated-signer receipts, and fail-closed experiment manifests make those boundaries auditable. The canonical receipt validates exact circuit and reference agreement, replay rejection, public-signal tamper rejection, and month-close reconciliation. It does not replace the four-dataset formal evaluation. **Future Work.** We will complete the preregistered S1–S5 server run, report target-host resource and failure distributions, evaluate attested

receiver hardware, and study release mechanisms that reduce settlement linkability without weakening auditability.

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